$ redroot scan --course pentest-pro
8 services found open

COURSE SYLLABUS

PROFESSIONAL PENETRATION TESTING

From your first Linux command to a full Red Team operation. Hands-on, live, and built on real-world engagements.

DURATION	MODULES	FORMAT	INSTRUCTOR
18 months	8	Live online + labs	Agampreet Singh

ABOUT THIS COURSE

This isn't a theory course. From the first month you'll be hacking real (legal) lab machines, using the same methods professional penetration testers use on real client engagements. By the end, you'll have the skills, the portfolio and the confidence to work in offensive security.

> **Before you start: your 30-minute placement interview**
>
> Before your batch starts, you'll have a 30-minute one-on-one interview with Agampreet. He'll ask about your current level and experience, and based on that you'll be placed in the batch that fits you best.

COURSE AT A GLANCE

0	Foundations	Months 1–2
1	OSINT & Reconnaissance	Month 3
2	Network & Active Directory Penetration Testing	Months 4–7
3	Web & API Penetration Testing	Months 8–11
4	Wireless Penetration Testing	Month 12
5	AI Penetration Testing	Months 13–14
6	Red Team Operations	Months 15–17
7	Capstone & Career Launch	Month 18

00

MONTHS 1–2

Foundations

“Every great hacker started here.”

Before you break systems, you need to know how they work. This module gives you the base that most self-taught hackers skip, and that later separates average pentesters from great ones.

You will learn to Move confidently in Linux and Windows, understand how data travels across a network, and write your own scripts to automate attacks.

- How networks really work: OSI/TCP-IP, IP addressing, ports, DNS, HTTP/HTTPS
- The Linux command line: files, permissions, processes, services
- Windows internals for hackers: PowerShell, users, services, registry
- Bash and Python scripting to automate your own hacking tools
- Building your personal hacking lab: Kali Linux, virtual machines, vulnerable targets
- How professionals work: methodology, scope, rules of engagement, and the law

Module project Build your own home lab and write your first Python port scanner

01

MONTH 3

OSINT & Reconnaissance

“The best hackers know everything about the target before they touch it.”

Real attacks start with information. You'll learn to map a company's entire attack surface using only public data.

You will learn to Find subdomains, exposed services, leaked credentials and employee information, then turn them into an attack plan.

- OSINT methodology and staying legal
- Subdomain and asset discovery
- Google dorking and search engine hacking
- Finding leaked credentials and exposed data
- Employee and email enumeration for social engineering
- Turning recon into an attack plan

Module project A complete recon report on a bug bounty target

02

MONTHS 4–7

Network & Active Directory Penetration Testing

“Break into networks the way real attackers do.”

This is the core of professional pentesting, and what most company engagements are about. You'll go from scanning a network to owning the entire Windows domain.

Prepares you for OSCP · PNPT · CRTP

You will learn to Find and exploit vulnerable services, escalate to root/SYSTEM, and compromise an Active Directory domain from start to finish.

- Host discovery, port scanning and service fingerprinting
- Vulnerability scanning and manual verification (no blind tool usage)
- Exploiting SMB, FTP, SSH, RDP, SNMP and more
- Metasploit and manual exploitation: understand what the exploit actually does
- Password attacks: hash cracking, brute force, credential reuse
- Linux and Windows privilege escalation
- Active Directory attacks: Kerberoasting, AS-REP roasting, pass-the-hash, BloodHound attack paths
- Pivoting and tunnelling into hidden network segments

Hands-on 20+ vulnerable lab machines, plus a multi-machine Active Directory lab

Module project Compromise a full corporate AD network and write a professional report

03

MONTHS 8–11

Web & API Penetration Testing

“Every company has a website. Most of them are vulnerable.”

Web apps are the most attacked surface in the world, and the biggest source of bug bounty rewards. You'll learn to find the bugs that scanners miss.

Prepares you for Burp Suite Certified Practitioner (BSCP) · eWPT · bug bounty programs

You will learn to Test any web application or API methodically, find critical vulnerabilities and prove their real impact.

- How web apps work: HTTP, sessions, cookies, same-origin policy
- A professional Burp Suite workflow
- The OWASP Top 10, in practice rather than theory
- Injection attacks: SQLi, XSS, SSTI, command injection
- Access control and logic bugs: IDOR, SSRF, authentication and session flaws
- File upload, path traversal, XXE and deserialization
- REST and GraphQL API hacking
- Bug bounty methodology and writing reports that get paid

Module project A full pentest of a vulnerable web app and API, plus a real bug bounty submission

04

MONTH 12

Wireless Penetration Testing

“If you can reach the Wi-Fi, you can reach the network.”

You'll learn to assess wireless networks and show how attackers get in from the parking lot.

Prepares you for OSWP

You will learn to Capture and crack Wi-Fi authentication, set up rogue access points and test enterprise wireless security.

- 802.11 fundamentals, monitor mode and packet capture
- WPA2 handshake and PMKID attacks
- WPA3 and its weaknesses
- Evil twin and rogue access point attacks
- Enterprise Wi-Fi (WPA2-Enterprise / EAP) testing

Module project A complete wireless security assessment

05

MONTHS 13–14

AI Penetration Testing

“The newest attack surface, and almost nobody knows how to test it yet.”

Every company is shipping AI chatbots and agents. Very few pentesters know how to break them. This module puts you ahead of most of the industry.

You will learn to Attack LLM-powered applications, bypass their guardrails, abuse AI agents and their tools, and report AI vulnerabilities professionally.

- How LLM apps, RAG pipelines and AI agents are built
- The OWASP Top 10 for LLM applications
- Direct and indirect prompt injection
- Jailbreaks and system prompt extraction
- Data leakage through RAG pipelines
- Abusing tools and plugins in AI agents
- Model attacks: data poisoning and evasion basics
- Reporting AI findings and recommending fixes

Module project Pentest a vulnerable AI chatbot and agent application

06

MONTHS 15–17

Red Team Operations

“Now you think like a real adversary.”

Everything comes together. Instead of finding every bug, you'll run objective-based operations: get in, stay hidden, and reach the crown jewels.

Prepares you for CRTO · CRTE · red team roles

You will learn to Plan and run a full red team engagement, from the first phishing email to domain dominance, while avoiding detection.

- Planning, threat emulation and rules of engagement
- Initial access and phishing simulation
- Command and control (C2) infrastructure
- Persistence, lateral movement and defense evasion
- Advanced Active Directory: golden/silver tickets, delegation abuse, domain persistence
- Purple teaming: working with defenders to improve detection
- Objective-based engagements from first access to final report

Module project A full red team operation against a simulated company

07

MONTH 18

Capstone & Career Launch

“Prove it. Then get hired.”

You'll run a complete penetration test on your own and walk away with a portfolio, a professional report and a clear path to your first (or next) job.

- A full end-to-end penetration test on a lab company
- Writing a client-ready pentest report
- Certification roadmap: OSCP, PNPT, CRTP, CRTO, BSCP: which to take and when
- Resume, LinkedIn, GitHub and portfolio review
- Mock technical interviews with Agampreet

Outcome Certificate of completion, plus a professional report and portfolio you can show employers

WHAT'S INCLUDED

- Live classes taught by Agampreet
- Hands-on lab work in every module
- Doubt-clearing support throughout the course
- Module projects to build your portfolio
- Certificate of completion

Questions? Email
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